

* Left Handed Weyl field ψ with Lagrangian:

$$\mathcal{L} = i \psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi = \frac{1}{2} m (\psi\psi + \psi^\dagger \psi^\dagger)$$

$\Rightarrow$ States both ψ and $\psi^\dagger$ are sep. particles with the same mass.

* Canonically conjugate momenta: $\pi^a \equiv \frac{\partial \mathcal{L}}{\partial \dot{\psi}_a(x)}$

$$\pi^a = i \psi^\dagger \bar{\sigma}^0 \dot{\psi}^a$$

Hamiltonian:

$$\begin{aligned} \mathcal{H} &= \pi^a \dot{\psi}_a - \mathcal{L} = i \psi^\dagger \bar{\sigma}^0 \dot{\psi}^a \dot{\psi}_a - i \psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi + \frac{1}{2} m (\psi\psi + \psi^\dagger \psi^\dagger) \\ &= i \psi^\dagger \bar{\sigma}^0 \dot{\psi}^a \dot{\psi}_a - i \psi^\dagger \bar{\sigma}^0 \partial_0 \psi - i \psi^\dagger \bar{\sigma}^i \partial_i \psi + \frac{1}{2} m (\psi\psi + \psi^\dagger \psi^\dagger) \\ &= -i \psi^\dagger \bar{\sigma}^i \partial_i \psi + \frac{1}{2} m (\psi\psi + \psi^\dagger \psi^\dagger) \end{aligned}$$

* Commutation relations:

$$\{\psi_a(\vec{x}, t), \psi_b(\vec{y}, t)\} = 0$$

$$\{\psi_a(\vec{x}, t), \pi^c(\vec{y}, t)\} = i \delta_a^c \delta^3(\vec{x} - \vec{y})$$

or

$$\{\psi_a(\vec{x}, t), i \psi^\dagger \bar{\sigma}^0 \dot{\psi}^c\} = i \delta_a^c \delta^3(\vec{x} - \vec{y})$$

$$\bar{\sigma}^0 = \sigma^0 = \mathbb{1}_{2 \times 2} \text{ and so}$$

$$(\sigma^0)_{\dot{\alpha}\beta} = \delta_{\dot{\alpha}\beta}$$

$$\delta_{\dot{\alpha}\beta} \{\psi_a(\vec{x}, t), \psi^\dagger_{\dot{\alpha}}(\vec{y}, t)\} = (\sigma^0)_{\dot{\alpha}\beta} \delta^3(\vec{x} - \vec{y})$$

$$\{\psi_a(\vec{x}, t), \psi^\dagger_{\dot{\alpha}}(\vec{y}, t)\} = (\sigma^0)_{\dot{\alpha}\beta} \delta^3(\vec{x} - \vec{y})$$

$$\{\psi^\alpha(\vec{x}, t), \psi^\dagger_{\dot{\alpha}}(\vec{y}, t)\} = (\sigma^0)^{\dot{\alpha}\beta} \delta^3(\vec{x} - \vec{y})$$

(Weyl fermion)

We can also write this in terms of a Dirac or Maj.

spinor.

* Let $\chi, \xi \in (1/2, 0)$, then $[1, 0]$

Dirac: $\Psi = \begin{pmatrix} \chi_c \\ \xi^\dagger_{\dot{c}} \end{pmatrix} \mid \bar{\Psi} \equiv \Psi^\dagger \beta \mid \beta \equiv \begin{pmatrix} 0 & \sigma^0 \\ 1 & 0 \end{pmatrix}$

$$\mathcal{L} = i\bar{\chi}^\dagger \bar{\sigma}^\mu \partial_\mu \chi + i\bar{\xi}^\dagger \bar{\sigma}^\mu \partial_\mu \xi - m(\chi\xi + \xi^\dagger\chi^\dagger) \quad \left(\begin{smallmatrix} \delta_a^c & 0 \\ 0 & \delta_a^c \end{smallmatrix} \right)$$

$$= i\bar{\Psi} \gamma^\mu \partial_\mu \Psi - m\bar{\Psi}\Psi \quad \bar{\Psi} = (\xi^\dagger, \chi^\dagger)$$

Claim: $\{\Psi_\alpha(\vec{x}, t), \Psi_\beta(\vec{y}, t)\} = 0$
 $\{\Psi_\alpha(\vec{x}, t), \bar{\Psi}_\beta(\vec{y}, t)\} = (\gamma^0)_{\alpha\beta} \delta^3(\vec{x} - \vec{y})$

Proof: $\alpha, \beta = 1, 2, 3, 4$

$$\left\{ \begin{pmatrix} \chi_c \\ \xi^{\dagger c} \end{pmatrix}, \begin{pmatrix} \xi^\alpha \\ \chi^\dagger_{\dot{\alpha}} \end{pmatrix} \right\} = \begin{pmatrix} \{\chi_c, \xi^\alpha\} & \{\chi_c, \chi^\dagger_{\dot{\alpha}}\} \\ \{\xi^{\dagger c}, \xi^\alpha\} & \{\xi^{\dagger c}, \chi^\dagger_{\dot{\alpha}}\} \end{pmatrix}$$

Cross terms vanish

$$\{\chi_c, \chi^\dagger_{\dot{\alpha}}\} = \sigma^0_{c\dot{\alpha}} \delta^3(\vec{x} - \vec{y})$$

$$\{\xi^{\dagger c}, \xi^\alpha\} = \{\xi^\alpha, \xi^{\dagger c}\} = \sigma^0_{\alpha\dot{c}} \delta^3(\vec{x} - \vec{y})$$

Plugging this in we have

$$\{\Psi_\alpha(\vec{x}, t), \bar{\Psi}_\beta(\vec{y}, t)\} = (\gamma^0)_{\alpha\beta} \delta^3(\vec{x} - \vec{y})$$

Majorana Field: $\Psi \equiv \begin{pmatrix} \psi_c \\ \psi^{\dagger\dot{c}} \end{pmatrix}, \bar{\Psi} = \bar{\Psi}^c C$

Recall $\Psi^c \equiv C \bar{\Psi}_m^T = \underline{\Psi}_m \quad (C^{-1} = C^T = -C)$

$$C^{-1} \underline{\Psi}_m = C^T \underline{\Psi}_m = \bar{\Psi}^T$$

$$\therefore \underline{\Psi}_m = \Psi^T C$$

$$C = \begin{pmatrix} -\epsilon^{ac} & 0 \\ 0 & -\epsilon^{\dot{a}\dot{c}} \end{pmatrix}$$

$$\mathcal{L} = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi - \frac{1}{2} m (\psi\psi + \psi^\dagger\psi^\dagger)$$

$$= \frac{i}{2} \bar{\Psi} \gamma^\mu \partial_\mu \Psi - \frac{1}{2} \bar{\Psi} \Psi$$

$$= \frac{i}{2} \Psi^T C \gamma^\mu \partial_\mu \Psi - \frac{1}{2} \Psi^T C \Psi$$

$$\{\Psi_\alpha(x, t), \Psi_\beta(y, t)\} = (C \gamma^0)_{\alpha\beta} \delta^3(\vec{x} - \vec{y})$$

$$\{\Psi_\alpha(x, t), \bar{\Psi}_\beta(y, t)\} = (\gamma^0)_{\alpha\beta} \delta^3(\vec{x} - \vec{y})$$

E.o.M: $\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \bar{\Psi})} \right) - \frac{\delta \mathcal{L}}{\delta \bar{\Psi}} = 0$

$$i/2 \gamma^\mu \partial_\mu \bar{\Psi} - \frac{1}{2} m \bar{\Psi} = 0$$

$$\not{d} \not{d} = -a^2$$

$$(i\not{\partial} - m)\bar{\Psi} = 0 \quad (\text{Dirac Eq.})$$

$\leadsto$ multiply by $(i\not{\partial} + m)(i\not{\partial} - m)\bar{\Psi} = 0$

$$i\cancel{\partial}(i\cancel{\partial} - m)\Psi + m(i\cancel{\partial})\Psi - m^2\Psi$$

$$-\cancel{\partial}\cancel{\partial}\Psi - im\cancel{\partial}\Psi + im\cancel{\partial}\Psi - m^2\Psi = 0$$

$$(\partial^2 - m^2)\Psi = 0 \quad \text{Klein-Gordon eq.}$$

$$\bullet (-\partial_t^2 + \partial_i^2 - m^2)\Psi = 0$$

$\leadsto$ Find a specific solⁿ: Sep of variables

$$(-\partial_t^2 + \partial_i^2 - m^2)T(t)X(x) = 0$$

$$0 = X(x)(-\partial_t^2 T(t)) + T(t)\nabla^2 X(x) - m^2 T(t)X(x)$$

$$\Rightarrow -\partial_t^2 T/T + \nabla^2 X/X - m^2 = 0$$

$$\nabla^2 X/X - m^2 = \lambda^2$$

$$-\partial_t^2 T/T = \lambda^2$$

$$-\partial_t^2 T = \lambda^2 T$$

$$T(t) = A e^{i\lambda t} + B e^{-i\lambda t}$$

$$\psi \rightarrow 0 \quad X \rightarrow \infty \quad A = 0$$

$X \dots$

$$\Psi(x) = u(\vec{p}) e^{ipx} + v(p) \bar{e}^{ipx}$$

$\Rightarrow$

$$(p + m)u(p) = 0$$

$$(-p + m)v(p) = 0$$

$$\Psi(x) = \sum_{s=\pm} \int d\vec{p} \left(b_s(p) u_s(p) e^{ipx} + d_s^\dagger(p) v_s(p) \bar{e}^{ipx} \right)$$